

Supporting Information for

Origin of Layered Herringbone Packing and Polymorphism in Polyacenes: A Quantum Chemical Optimization Approach

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1. Crude Search Using the vdW Contact Model

In the first step of stepwise optimization, we employed a van der Waals (vdW) contact model to roughly optimize molecular positions and orientations prior to fine-tuning based on quantum chemical calculations of intermolecular interactions. In this model, all atoms are treated as rigid spheres defined by their vdW radii. The procedure is illustrated in [Fig. S1](#). Lattice parameters (a and b) and molecular orientation angle (α) were optimized to minimize the unit cell area while avoiding overlap between the vdW spheres of atoms in neighboring molecules. Minimizing the cell area is advantageous because it enhances dispersion interactions, thereby lowering the interaction energy. In addition, avoiding vdW sphere overlap is essential due to the repulsive nature of exchange interactions. A similar procedure was applied for the crude search of interlayer stacking arrangements.

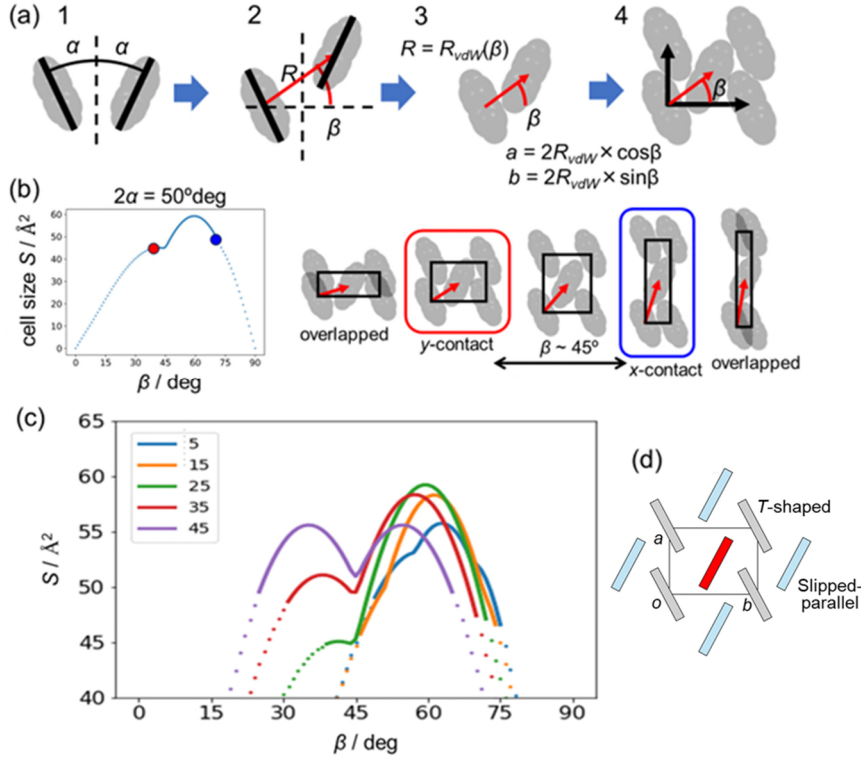


Fig. S1. Geometric search for initial molecular arrangements. (a) Schematic illustration of the geometric search procedure used to generate initial molecular arrangements at each α . (1) A basic molecular pair is constructed with an angle α using a glide symmetry operation. (2) A polar coordinate system (R, β) is introduced to describe the relative configuration of the pair. (3) The van der Waals (vdW) contact distance, R_{vdW} , is determined for each β . (4) Additional molecules are generated via translational operations to form a monomolecular layer, with the intralayer lattice constants a and b defined for each β as: $a = 2 R_{vdW} \cos \beta$, and $b = 2 R_{vdW} \sin \beta$. (b) Plot of unit cell area ($S = a \times b$) as a function of β for $2\alpha = 50^\circ$. Solid curves indicate that the interatomic distances between adjacent slipped-parallel molecules exceed the vdW contact distances, while dashed curves indicate overlapping between adjacent slipped-parallel molecules. Initial molecular arrangements satisfying the vdW contact condition are marked with red and blue dots, and the corresponding structures are shown in the right panel. (c) Similar plots of S versus β for various values of α (5° , 15° , 25° , 35° , and 45°), showing that the local minimum at $\beta = 45^\circ$ becomes more pronounced as α approaches 45° . (d) Illustration of the eight nearest-neighbor interactions within the intralayer structure, consisting of four T-shaped and four slipped-parallel contacts.

2. Precision Search Using Energy Minimization

We calculated the total intermolecular interaction energies within both the molecular layer and the full crystal using DFT-D calculations, and examined how these energies vary with changes in configuration parameters. For each optimization step, we defined appropriate calculation ranges for intralayer interactions ($E_{\text{intra}(6)}$, $E_{\text{intra}(8)}$), interlayer interactions ($E_{\text{inter}(7)}$), and the entire crystal ($E_{\text{int(near)}}$, $E_{\text{int(G1)}}$, $E_{\text{int(G2)}}$, $E_{\text{int(all)}}$). We then computed the sum of the interaction energies for the initial molecular arrangements generated by the vdW contact model, as well as for trial arrangements obtained by slightly perturbing each configuration parameter.

The most stable arrangement—corresponding to the lowest total interaction energy—was adopted as the new starting point for the next optimization step. By repeating this iterative process, we identified locally stable arrangements corresponding to energy minima. The variation in total interaction energy was plotted as a function of each configuration parameter to construct potential energy maps.

Figure S2 illustrates the optimization of the molecular layer during the first step. In addition, comprehensive energy potential maps of $E_{\text{intra}(8)}$ were calculated, confirming that our optimization covered all locally stable structures.

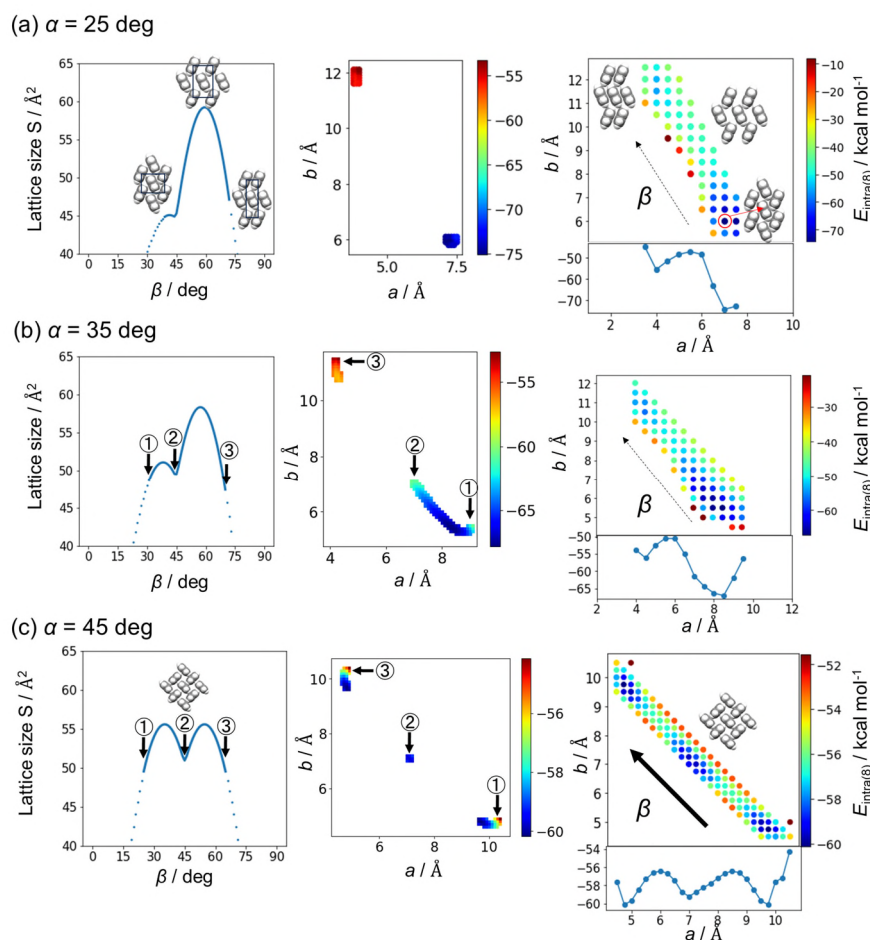


Fig. S2. Minimization of the sum of intermolecular interaction energies for pentacene monomolecular layers. (a–c) Left panels: Geometric search for initial molecular arrangements. Plots of unit cell area ($S = a \times b$) as a function of β . Center panels: Lattice energy minimization results using the hill-climbing approach. Right panels: Potential energy maps of $E_{\text{intra}(8)}$ (limited to values < -40 kcal/mol), highlighting regions of favorable molecular arrangements. Each lower right panel shows $E_{\text{intra}(8)}$ plotted as functions of a along the local energy valleys indicated in the corresponding upper panel. These calculations were performed for the pentacene monomolecular layer at (a) $\alpha = 25^\circ$, (b) 35° , and (c) 45° , illustrating how $E_{\text{intra}(8)}$ varies with molecular arrangement across different α values. In the calculation of $E_{\text{intra}(8)}$, three of the eight intermolecular interactions (one T-shaped and two slipped-parallel contacts) are independent due to the molecular symmetry of the polyacene backbone and the symmetry of the arrangements.

3. Variation of Intralayer Packing on α and n

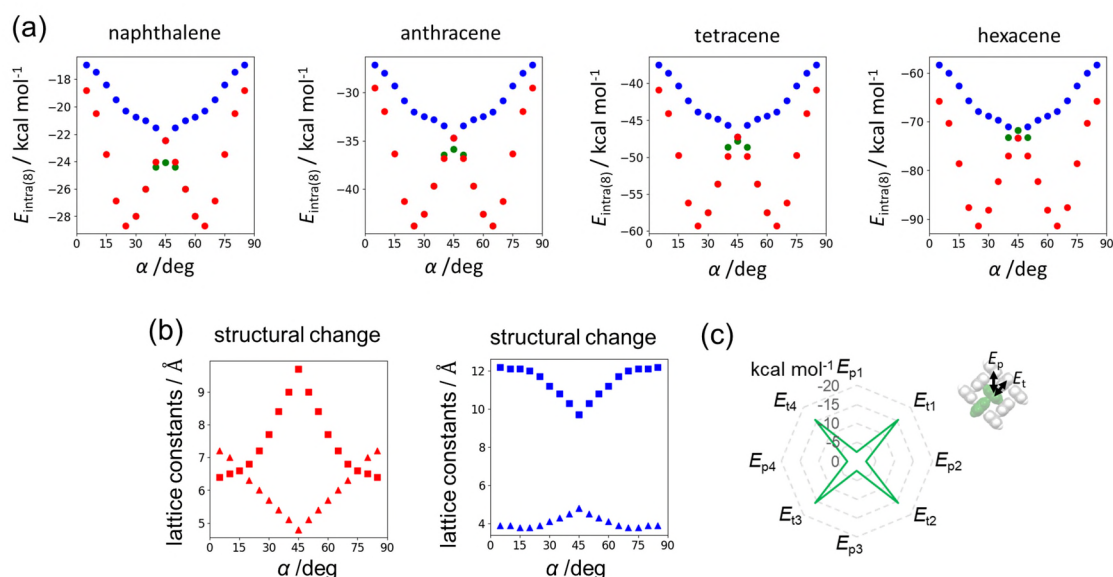


Fig. S3. (a) Similar plots to Fig. 2b, which was for pentacene, are shown here for naphthalene, anthracene, tetracene, and hexacene. The intralayer interaction energy $E_{\text{intra}(8)}$ of stable and metastable molecular arrangements is plotted as a function of α . The lattice parameters a and b were optimized for each value of α . HB, PS, and CH packing motifs are shown in red, blue, and green, respectively. (b) Optimized lattice constants a and b for HB (left panel) and PS (right panel) packings of pentacene as a function of α . Triangles and squares indicate the intermolecular distances between the directly contacted slipped-parallel pairs and the distances between the second-nearest slipped-parallel pairs, respectively. (c) Radar chart showing the calculated intermolecular interactions among eight adjacent molecules in the CH packing of pentacene.

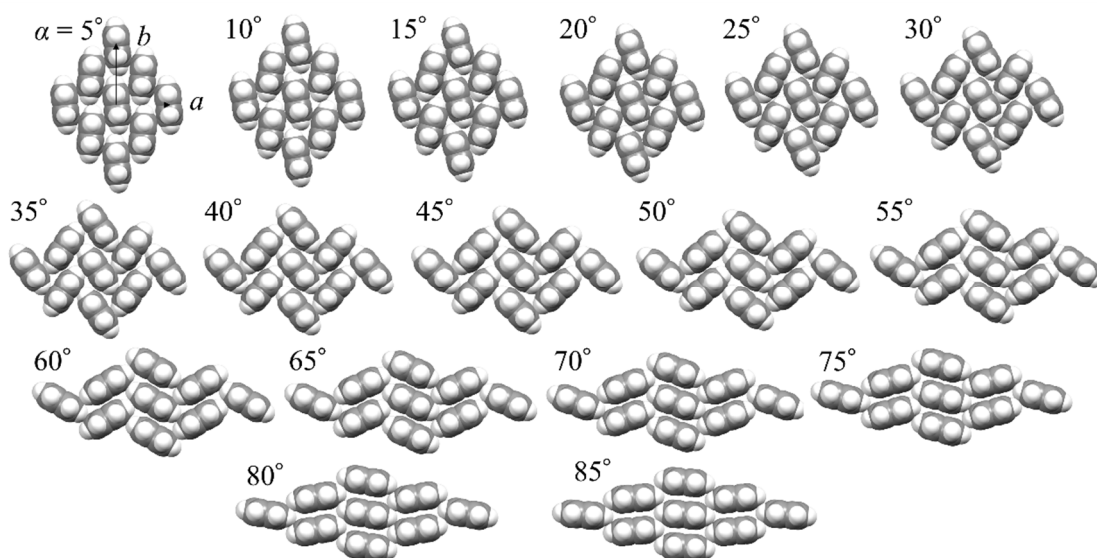


Fig. S4. Comprehensive view of stable (shown by red filled circles at $\alpha < 45^\circ$ in Fig. 2(b)) and metastable (blue filled circles at $45^\circ < \alpha$ in Fig. 2(b)) intralayer molecular arrangements optimized at each α for pentacene, viewed along the molecular long axes.

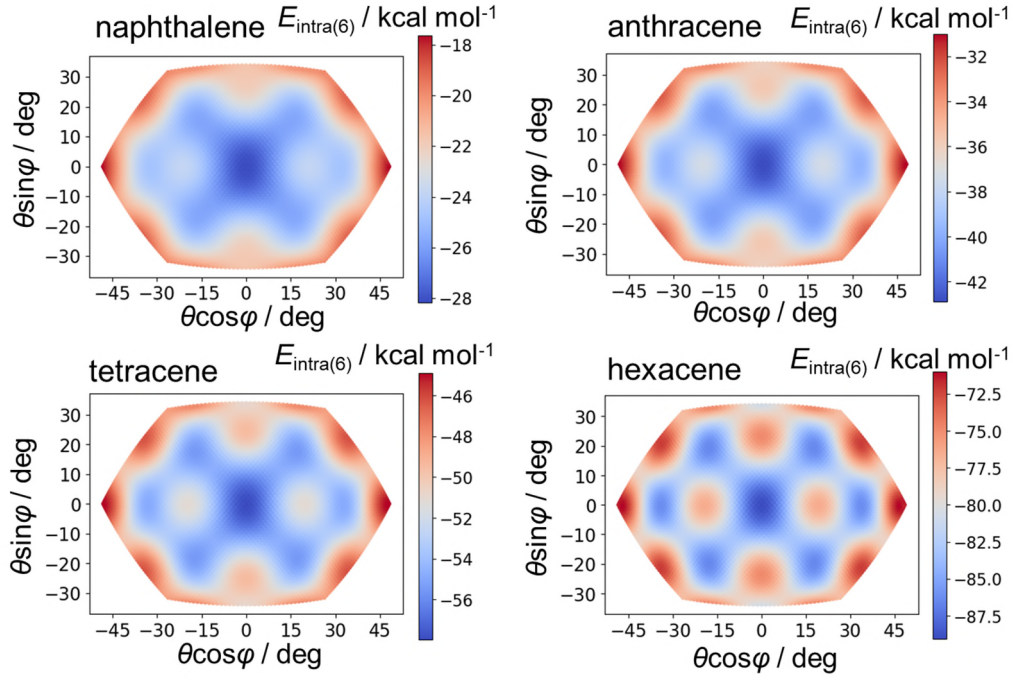


Fig. S5. Similar plots with Fig. 5b for pentacene. $E_{\text{intra}(6)}$ landscape for the monomolecular layers of (a) naphthalene, (b) anthracene, (c) tetracene, and (d) hexacene, with varying φ_{incl} and θ_{incl} .

4. Actual Treatment of Uniform Long-Axis Inclination

A uniform long-axis inclination of molecules in the R -form monomolecular layer is actually treated by an operation of long-axis sliding shift ΔZ_t and ΔZ_p in the unit cell. Then the uniform long-axis inclination is presented by two variables; θ_{incl} and φ_{incl} . These are related with ΔZ_t and ΔZ_p by the following equations.

$$\theta_{\text{incl}} = \cos^{-1} \left\{ 1 / \sqrt{1 + ((2Z_t - Z_p)/a)^2 + (Z_p/b)^2} \right\}$$

$$\varphi_{\text{incl}} = \arg\{(2Z_t - Z_p)/a + i Z_p/b\}$$

Conversely, ΔZ_t and ΔZ_p are related with θ_{incl} and φ_{incl} by the following equations.

$$Z_t = \{\text{sgn}(\cos \varphi_{\text{incl}}) \times a + \text{sgn}(\sin \varphi_{\text{incl}}) \times b |\tan \varphi_{\text{incl}}|\} \frac{\tan \theta_{\text{incl}}}{2\sqrt{1 + \tan^2 \varphi_{\text{incl}}}}$$

$$Z_p = \text{sgn}(\sin \varphi_{\text{incl}}) \times b |\tan \varphi_{\text{incl}}| \frac{\tan \theta_{\text{incl}}}{\sqrt{1 + \tan^2 \varphi_{\text{incl}}}}$$

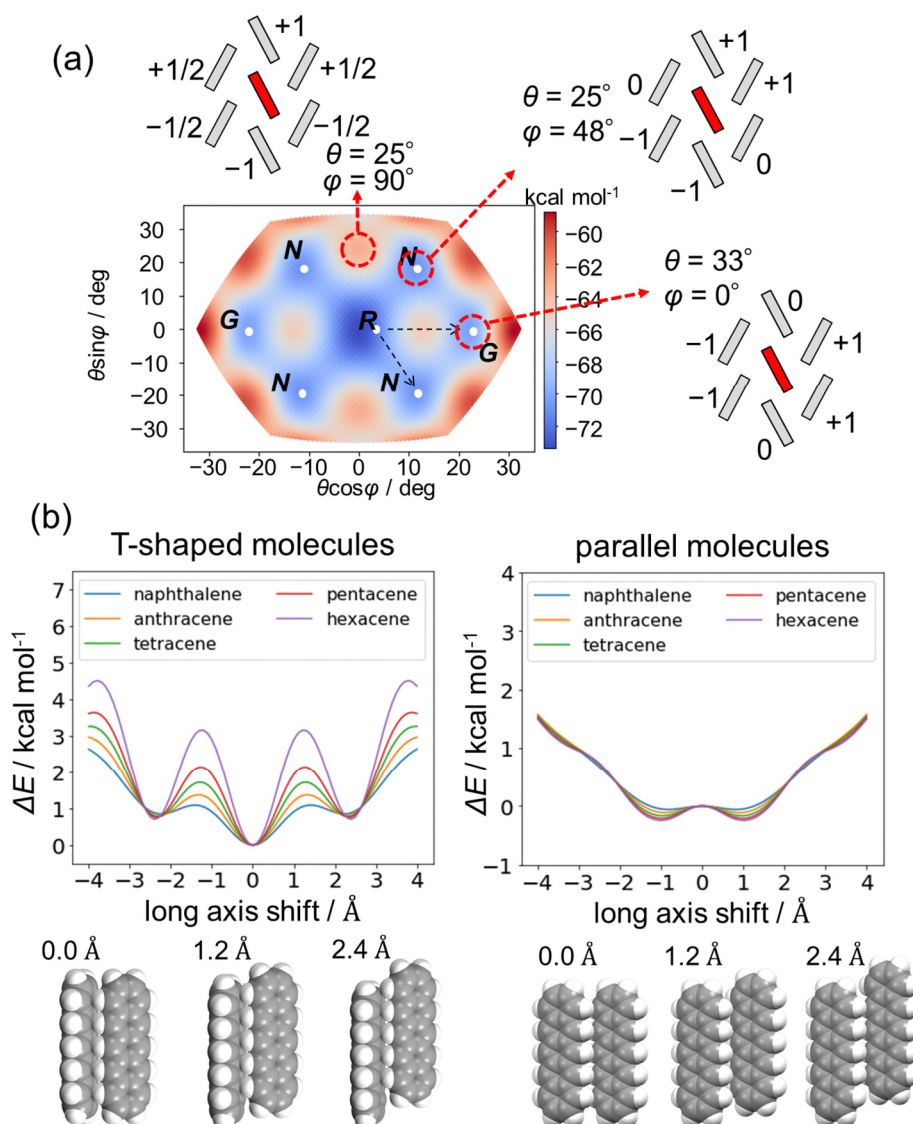


Fig. S6. (a) Schematic representation of the relative long-axis sliding between six neighboring molecules in the locally stable *N* and *G* forms, as well as in an unstable arrangement at $\varphi_{\text{incl}} = 90^\circ$ and $\theta_{\text{incl}} \approx 0^\circ$ in pentacene. The potential map is the same as in Fig. 5(b). In the *N* form, a one-ring-length misalignment occurs at the *T*-shaped contacts between two adjacent molecules, while in the *G* form, the same misalignment occurs between four adjacent molecules. (b) Variation in interaction energy as a function of long-axis sliding at *T*-shaped contacts (left panel) and slipped-parallel contacts (right panel).

5. Optimization of Interlayer Stacking Arrangements

To optimize the interlayer stacking arrangements, it is essential to determine the relative positions of molecules in adjacent layers while considering the intralayer HB arrangement. Figure S7 illustrates the relationship between a molecular pair in a *T*-shaped contact and the surrounding molecules in the neighboring layer. For each molecule, the seven nearest neighbors in the adjacent layer were considered. Due to translational

symmetry, four of these pairs exhibit equivalent positional relationships (as shown in the lower part of the figure), resulting in ten unique molecular pairs used to calculate the interlayer interaction energy, $E_{\text{inter}(7)}$.

In the Type-III structure, which preserves glide symmetry, five of these molecular pairs are symmetry-equivalent. As a result, the interlayer interaction energy is calculated from five unique molecular pairs: two *T*-shaped contacts and three slipped-parallel contacts.

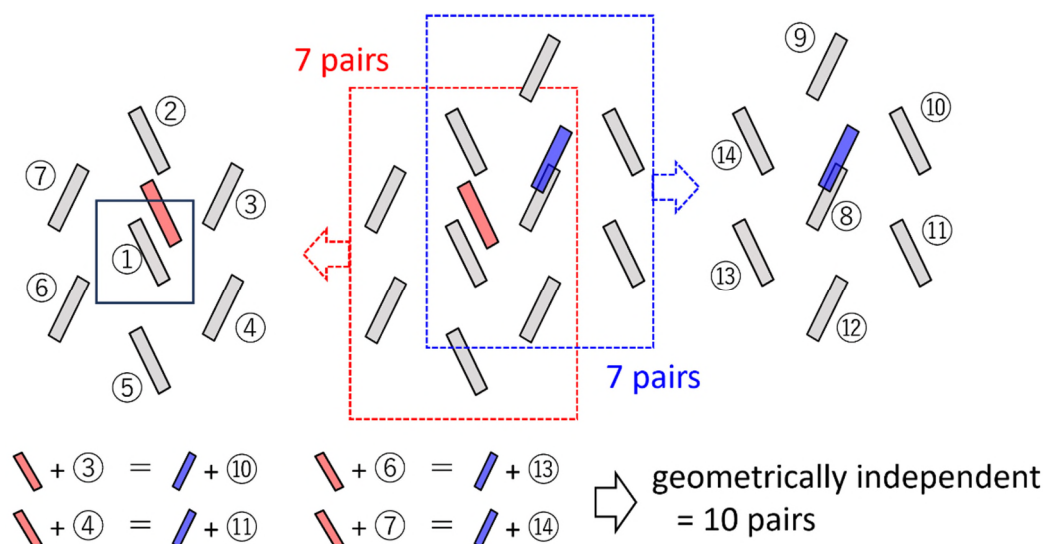


Fig. S7. Relationship between a molecular pair in a *T*-shaped contact and the surrounding molecules in adjacent molecular layers. Center: Relative positions of a molecular pair in a *T*-shaped contact and the surrounding molecules in adjacent layer. Left: The central molecule (red) and its seven nearest neighbors in the adjacent layer (the closest molecule and its six surrounding neighbors), viewed along the molecular long axis. Right: The molecule forming the *T*-shaped contact (blue) and its seven nearest neighbors in the adjacent layer (the closest molecule and its six surrounding neighbors), also viewed along the molecular long axis.

6. Calculation of Total Intermolecular Interactions in Crystals

The intermolecular interaction energy between neutral polyacene molecules decreases rapidly with increasing distance, as it is primarily governed by a balance between attractive dispersion forces and repulsive exchange-repulsion forces. Therefore, when optimizing each stable polymorph, it is sufficient to consider only the interactions between adjacent molecules. However, to compare the energies of the resulting polymorphic structures, it is both useful and necessary to evaluate the total interaction energy—including contributions from distant molecules—using an extrapolation method.

To achieve this, we systematically classified the molecules surrounding a central molecule. The position of each molecule is expressed as a linear combination of the lattice vectors, where the coefficients are integers or half-integers within our defined parameter space. Taking advantage of the periodicity of the crystal lattice, we

classified molecules based on their L1 norm—that is, the sum of the absolute values of the coefficients in the lattice vector representation of the displacement between molecules. The coordinates of the type X molecules are given by:

$$V = la + mb + nc$$

where l and m are integers or half-integers, n is an integer, and a and b are the intralayer lattice vectors, while c is the interlayer lattice vector. The classification criteria are as follows:

1. $|l| + |m| = X$ and $n = 0$ (intralayer molecules),
2. $|l| + |m| + |n| = X + 1$ (for $|n| \neq 0$ and $|l| + |m| > 0$; interlayer molecules),
3. $|l| + |m| = 0$ and $|n| = X$ (molecules directly above or below the central molecule).

This method enables the systematic classification of molecules in the crystal according to their intermolecular distances, by leveraging the discrete and periodic nature of molecular positions, as illustrated in Fig. S8.

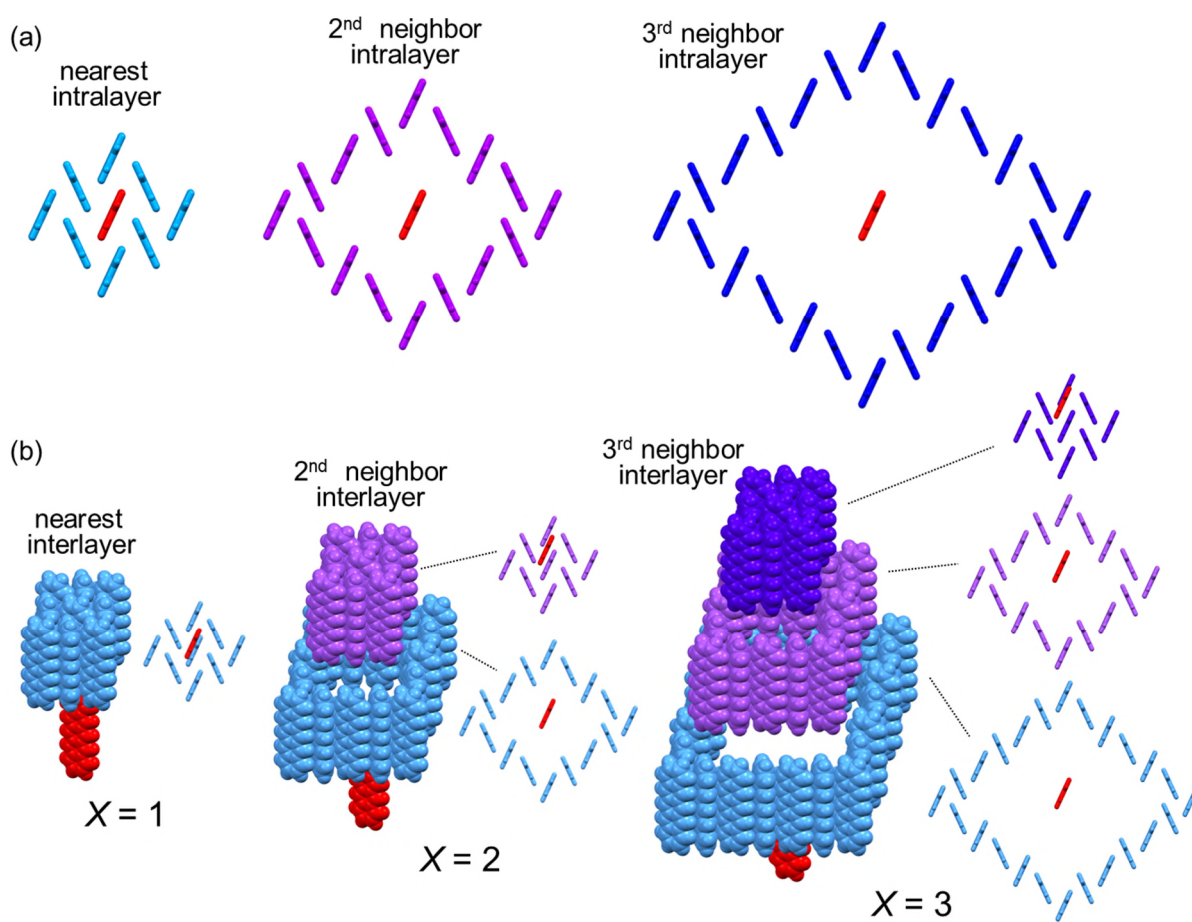


Fig. S8. (a) Schematic illustration of molecular groups within the same layer—nearest neighbors, second-nearest neighbors, and third-nearest neighbors—shown in light blue, purple, and blue, respectively, and viewed along the molecular long axis. (b) Schematic illustration of molecular groups spanning both the same and adjacent layers, including nearest neighbors, second-nearest neighbors, and third-nearest neighbors. Molecules in the nearest-neighbor layer, second-nearest-neighbor layer, and third-nearest-neighbor layer are shown in light blue, purple, and blue, respectively.

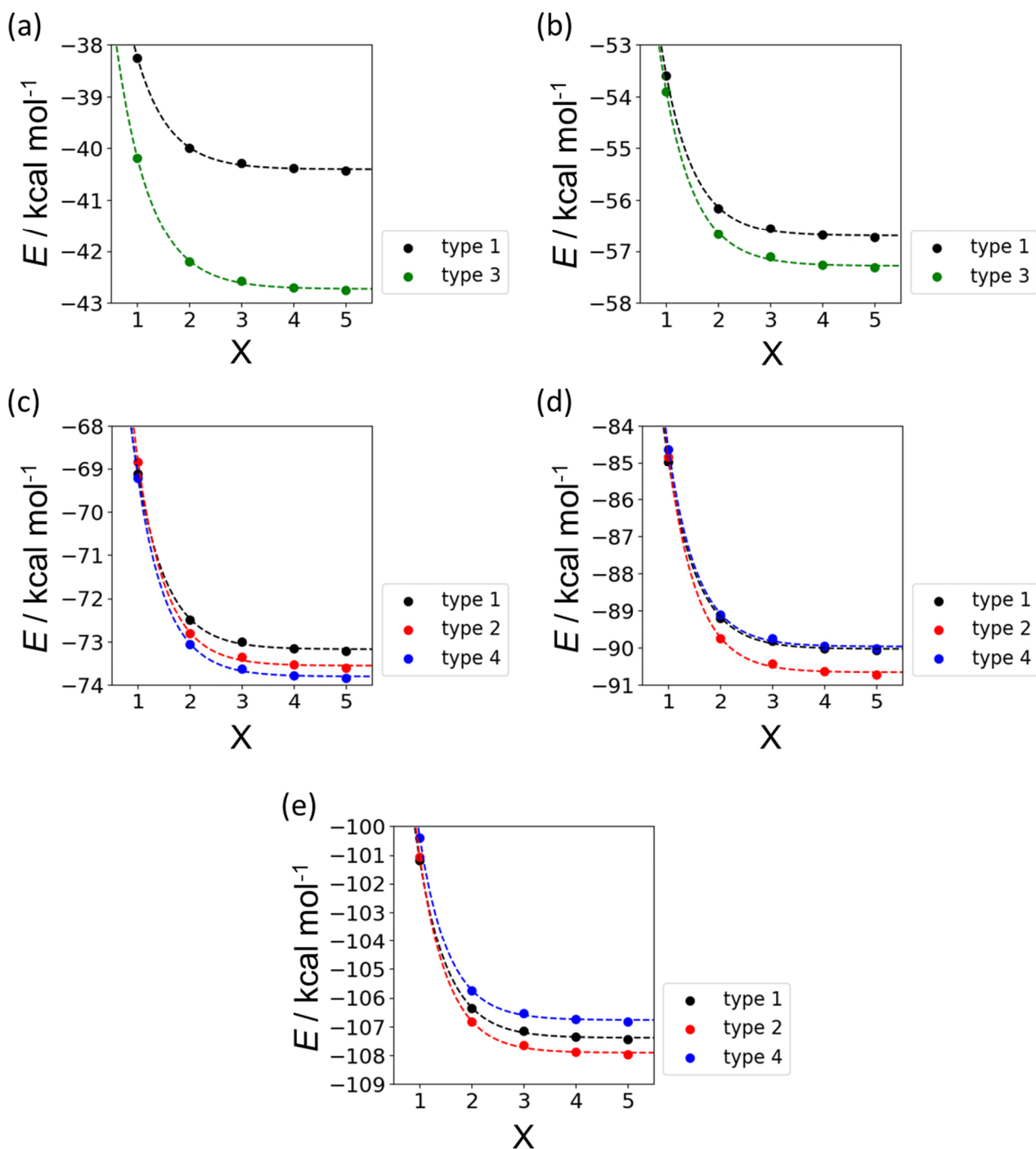


Fig. S9. Plots of the sums of intermolecular interaction energies as a function of X , the classification index representing the distance between the central molecule and its surrounding molecules as depicted in Fig. S8, for (a) naphthalene, (b) anthracene, (c) tetracene, (d) pentacene, and (e) hexacene. Dashed curves indicate the fitting exponential curves used to estimate the convergence behavior and total intermolecular interactions across the entire crystal.

7. Comparison with Experimental Structures

To compare the optimized molecular arrangements with experimental crystal structures, we extracted molecular displacement parameters from the crystallographic data. First, the molecular long axis and molecular plane were determined using principal component analysis. The interplanar dihedral angle between molecules in a *T*-shaped contact was calculated from the dot product of their normal vectors. In cases where the molecular long axes are aligned in parallel, the molecular shift was evaluated using the dot product of the unit vector along the molecular long axis and the translation vector.

We also simulated powder X-ray diffraction (XRD) patterns based on the optimized model structures for comparison with the experimental data. The reciprocal lattice was constructed from the optimized lattice vectors, and for each scattering vector, the diffraction angle, atom scattering factor, and structure factor were calculated. To facilitate comparison of diffraction angles and peak intensities, Laue's function with $N = 10,000$ was employed. Additionally, the Lorentz-polarization (LP) factor, given by $(1 + \cos^2 2\theta) / \sin^2 \theta \cos \theta$, was included.

Table S1. Optimized parameter values for reproducing the Type III structures of naphthalene and anthracene using stepwise optimization calculations. The final parameter values were refined by applying a long-axis twist operation under glide symmetry (see Sect. 2.4 and Fig. 7). These values are compared with those extracted from the experimental crystal structure analyses.

Note: The parameters a and b here do not correspond to the crystallographic lattice constants. The consistency between the lattice parameters and molecular orientations in the experimental crystal structure and the optimized models is further discussed through the comparison of simulated and experimental powder XRD pattern in Fig. 9.

		$a / \text{\AA}$	$b / \text{\AA}$	α / deg	$\theta_{\text{twist}} / \text{deg}$	$\Delta Z_T / \text{\AA}$	$x / \text{\AA}$	$y / \text{\AA}$	$z / \text{\AA}$
Naphthalene	exp	5.9	7.4	65	13	1.5	0.0	1.7	8.4
Type III	calc	5.8	7.5	65	13	1.6	0.0	1.6	8.4
Anthracene	exp	6.0	7.5	65	8.3	1.9	0.0	1.5	11.0
Type III	calc	5.9	7.5	65	8.0	2.0	0.0	1.5	10.9

Table S2. Optimized parameter values for reproducing the Type IV structures of tetracene; the Type I, II, and IV structures of pentacene; and the Type II structure of hexacene, via stepwise structure optimization. The parameters for the Type I structure were obtained based on a regular intralayer molecular arrangement (Sect. 2.1 and Fig. 2). The Type II structure parameters were refined using uniform long-axis inclination (see Sect. 2.2 and Fig. 5), while those for the Type IV structure were optimized through non-uniform long-axis inclination (see Sect. 2.5 and Fig. 8). These optimized values are compared with those extracted from experimental crystal structure analyses.

		$a / \text{\AA}$	$b / \text{\AA}$	α / deg	$\theta_{\text{incl}} / \text{deg}$	$\varphi_{\text{incl}} / \text{deg}$	$\theta'_{\text{incl}} / \text{deg}$	$x / \text{\AA}$	$y / \text{\AA}$	$z / \text{\AA}$
Tetracene	exp	7.5	5.8	26	24	43	2.8	1.4	-1.2	13.3
Type IV	calc	7.4	5.7	25	20	46	2.2	1.6	-1.1	13.1
Pentacene	exp	7.7	5.9	27	22	41	1.9	1.5	-1.0	15.9
Type IV	calc	7.4	5.8	25	22	43	1.3	1.5	-1.0	15.6
Pentacene	exp	7.6	5.9	27	6.0	-75	0.6	2.1	0.6	15.5
Type I	calc	7.2	6.0	25	0.0	—	0.0	1.9	0.0	15.4
Pentacene	exp	7.4	5.9	26	25	48	0.8	0.3	1.4	16.0
Type II	calc	7.2	5.9	25	27	48	0.3	0.2	1.3	16.0
Hexacene	exp	7.3	5.9	26	26	47	0.6	0.2	1.4	18.5
Type II	calc	7.2	5.9	25	27	48	0.3	0.2	1.3	18.5